Topology weakens Monotonicity via trade-off(s)

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1 Setup:

Let B be a topological space Consider outcome relations on B of the form:
pRU, $U \subseteq C \subset \overline{U} \Rightarrow pRC$ (Weak Monotonicty: Monotonicity upto closure)

2 Algebraic structures, Concrete and Abstract

Fix a topological space B Form algebra with pairs (R S) where R S are outcome relations satisfying weak monotonicity and consistency. The operators $\cup, \cap, -, \circ$ and their definitions do not change.

Example:
B = ($\mathbb{R}$, Euclidean Topology).
Let R = $\{(p, [0, 1]) \mid p \in \mathbb{R}\}$.

Finally R is weakly (not fully) monotonic and (R,R) are consistent with each other.

$(R, R) \cup (R, R) = (R \cup R, R \cap R) = (R, R)$
$(R, R) \cap (R, R) = (R \cap R, R \cup R) = (R, R)$
$(R, R)^{-} = (R, R)$
$(R, R) \circ (R, R) = (R \circ R, R \circ R)$

Call this class of algebras locally-monotone-algebras.

The axioms G1-G11 are satisfied except G8.

3 Toolkit

This time in the definition of prime filters, we exclude the empty filter. Venema doesn't, however nothing in his proofs will change if empty is excluded.

$\hat{a}, F_T$ do not change.

The reader can verify that $\hat{a}$ form basic closed sets.

Now given a distributive lattice D with a monotone map $\diamond$, form outcome relations on the topological space of prime filters as follows:

(1) First do it for me open sets U:
$pQ_{\Diamond}U \iff \diamond a \in p$ for all $a \in F_T$.
(2) Then do for closed sets as above.
(3) Extend all open sets upto closure.

Lemma:

$$pQ_{\Diamond}\hat{a} \text{ iff } \Diamond a \in p.$$

4 Main Theorems:

No change in 4.1 except composition which stands unresolved. 4.2, 5.1 remain as they are.

Consistency:
As before set $B' = B \cup \infty$,

Set Topology as: T(B)$\cup\{\{\infty\} \cup (B \setminus K) : K \subseteq B$ is compact$\}$.

For R over B, define:
$R' = \{(\infty, T) \mid \infty \in T\} \cup \{(p, T) \mid p \in B, \infty \in T, (p, T^-) \in R\}$

Consistency easily verified.

Let us verify, R is weakly monotonic:
Let $(p, C) \in R'$ and $U \subseteq C \subseteq \overline{U}$.
If $p = \infty$, then $\infty \in U$
So, $\infty \in C$.

$$\therefore (p, C) \in R'.$$

Now let $p \neq \infty$, $(p, U) \in R'$, $U \subseteq C \subseteq \overline{U}$,

$$(p, U^-) \in R$$

$$\therefore U^- \subseteq C^- \subseteq \overline{U}^-$$

$$\overline{U}^- = (\overline{U}^-) \quad \text{(This is true)}$$

$$\therefore (p, C^-) \in R$$

$$\therefore (p, C) \in R'$$